Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A bar magnet is cut into 3 pieces. Does this produce isolated north and south poles? Sketch the usual result.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp. Label each input and useful output.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A hand-cranked generator receives 50 W of mechanical power and supplies 12 V at 3 A. Complete the energy-rate account. Find electrical output power, efficiency and the rate of transfer to other forms.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

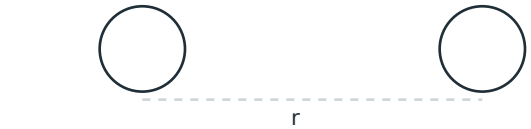
05 Explain your reasoning

A motor drives a generator that feeds the motor. Explain why real losses prevent the loop from running forever without external input.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Move from r to 4 r away from a point charge. Compare field magnitude and state the source model assumption. Label the source charge, distances and electric-field directions.



Marzano check - circle: 1 2 3

07 Exit check

A motor is 80% efficient and a generator is 75% efficient. Find their combined useful fraction in sequence.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A motor receives 24 W and operates at 75% efficiency. It lifts a 3 kg load at steady speed with g=10. Find the speed and wasted power.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3